1. Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).

```
KD3-Prerana-96192>SELECT * FROM orders WHERE cnum = (SELECT cnum FROM customers WHERE cname = 'Cisneros'
);
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+------+---------+------------+------+------+
2 rows in set (0.00 sec)
```

2. Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table.

```
KD3-Prerana-96192>SELECT o.amt, c.cname, c.rating FROM orders o INNER JOIN customers c ON o.cnum = c.cnu
m WHERE o.amt > (SELECT AVG(amt) FROM orders);
+---------+---------+--------+
| amt     | cname   | rating |
+---------+---------+--------+
| 5160.45 | Liu     |    200 |
| 9891.88 | Clemens |    100 |
| 4723.00 | Clemens |    100 |
+---------+---------+--------+
3 rows in set (0.00 sec)
```

3. Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the

table.

```
KD3-Prerana-96192>SELECT snum, SUM(amt) AS Total FROM orders GROUP BY snum HAVING SUM(amt) > (SELECT MAX
(amt) FROM orders);
+------+----------+
| snum | Total    |
+------+----------+
| 1001 | 15382.07 |
+------+----------+
1 row in set (0.00 sec)
```

4. Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'. (Hint: find all customers of serres salesperson).

```
KD3-Prerana-96192>SELECT cname, rating FROM customers WHERE rating >= ANY(SELECT rating FROM customers W
HERE snum = (SELECT snum FROM salespeople WHERE sname= 'Serres'));
+----------+--------+
| cname    | rating |
+----------+--------+
| Giovanni |    200 |
| Liu      |    200 |
| Grass    |    300 |
| Cisneros |    300 |
+----------+--------+
4 rows in set (0.00 sec)
```

5. Write a query that will find all salespeople who have no customers located in their city. (Hint: Choose the correct keyword ANY/ALL).

```
KD3-Prerana-96192>SELECT * FROM salespeople s WHERE s.city <> ALL(SELECT c.city from customers c WHERE c.snum= s.snum
);
+------+---------+-----------+------+
| snum | sname   | city      | comm |
+------+---------+-----------+------+
| 1004 | Motika  | London    | 0.11 |
| 1007 | Rifkin  | Barcelona | 0.15 |
| 1003 | Axelrod | New York  | 0.10 |
+------+---------+-----------+------+
3 rows in set (0.00 sec)
```

6. Write a query that selects all orders for amounts greater than any for the customers in London.

```
KD3-Prerana-96192>SELECT * FROM orders WHERE amt > ANY(SELECT amt FROM orders WHERE cnum IN (SELECT cnum FROM customers WHERE city =
'london'));
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
6 rows in set (0.00 sec)
```

7. Extract all the orders of Motika.

```
KD3-Prerana-96192>SELECT * FROM orders WHERE snum = (SELECT snum FROM salespeople WHERE sname = 'Motika')
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+------+---------+------------+------+------+
1 row in set (0.00 sec)
```

8. Find all the orders of salespeople servicing customers in London.

```
KD3-Prerana-96192>SELECT * FROM orders WHERE snum IN (SELECT snum FROM customers WHERE city = 'London');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
3 rows in set (0.00 sec)
```

9. Find names and numbers of all salesperson who have more than one customer.

```
KD3-Prerana-96192>SELECT sname, snum FROM salespeople WHERE snum IN (SELECT snum FROM customers GROUP BY snum HAVING COUNT(snum)>1 );

+--------+------+
| sname  | snum |
+--------+------+
| Peel   | 1001 |
| Serres | 1002 |
+--------+------+
2 rows in set (0.00 sec)
```

10. Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```
KD3-Prerana-96192>SELECT s.snum, s.sname, s.city, (SELECT COUNT(*) FROM customers c WHERE c.snum = s.snu
m) as cnt FROM salespeople s WHERE (SELECT COUNT(*) FROM customers c WHERE c.snum = s.snum) > 1;
+------+--------+----------+------+
| snum | sname  | city     | cnt  |
+------+--------+----------+------+
| 1001 | Peel   | London   |    2 |
| 1002 | Serres | San Jose |    2 |
+------+--------+----------+------+
2 rows in set (0.00 sec)
```

11. Select customers who have a greater rating than any other customer in Rome.

```
KD3-Prerana-96192>SELECT * FROM  customers WHERE rating >= ANY(SELECT MAX(rating) FROM customers WHERE c
ity = 'Rome');
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.00 sec)
```

12. Select all orders that had amounts that were greater that at least one of the orders from ‘1990-10-04’.

```
KD3-Prerana-96192>SELECT * FROM orders WHERE amt > (SELECT MIN(amt) FROM orders WHERE odate = '1990-10-04');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
8 rows in set (0.00 sec)
```

13. Find all orders with amounts smaller than any amount for a customer in San Jose.

```
KD3-Prerana-96192>SELECT * FROM orders WHERE amt < ANY(SELECT amt FROM orders where cnum IN (SELECT cnum from customers where  city =
 'San Jose'));
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 |   75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
+------+---------+------------+------+------+
8 rows in set (0.00 sec)
```

14. Select those customers whose rating are higher than every customer in London.

```
KD3-Prerana-96192>SELECT * FROM customers WHERE rating > ALL (SELECT rating FROM customers WHERE cnum IN (SELECT cnum FROM customers
WHERE city = 'London'));
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.00 sec)

KD3-Prerana-96192>
```